

**STEM Course Prep**  
**In Class Activities**  
**Chapter 6**

*Slides 2-6*

1. Create a model to analyze the velocity and displacement of a ship at sea. Use a function to calculate velocity and displacement of the ship for a time range of 0-60 minutes. Plot the results.

$$\text{displacement} = -5t^2 + 90t - 80$$

$$\text{velocity} = 90 - 10t$$

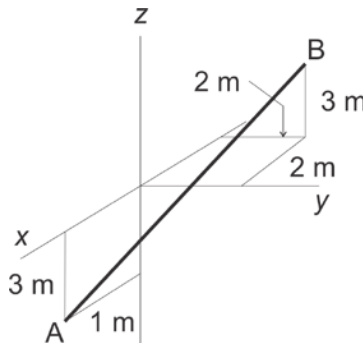
Script file

```
time=0:.5:60;  
[displace,velocity]=ship(time)  
subplot(1,2,1)  
plot(time,displace)  
xlabel('time (min)')  
ylabel('displacement (ft)')  
subplot(1,2,2)  
plot(time,velocity)  
xlabel('time (min)')  
ylabel('velocity (ft/s)')
```

ship.m

```
function [ s, v ] = ship( t )  
%calculates the velocity and displacement  
s=-5*t.^2+90*t-80;  
v=90-10*t;  
end
```

2. Create a function that will allow the user to find the length between two points, the unit vector of that line and the line's direction. Use the problem shown below for the test case.



Script file

```
ptA=[1,0,-3];  
ptB=[-2,2,3];  
[r,u,alpha,beta,gamma]=statics(ptA,ptB)
```

statics.m

```
function [ r,uv,alp,bet,gam ] = statics( A,B )  
%This function will calculate the distance between two points (A and B), the unit  
%vector of a line going from A to B, and the cosine direction angles  
r_vec=B-A;  
r=norm(r_vec);  
uv=r_vec/r;  
alp=acosd(uv(1));  
bet=acosd(uv(2));  
gam=acosd(uv(3));  
end
```

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*Slides 7-14*

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3. Create a function that will plot the cosine of the given angle. In the main script, call that function and then plot a sine function on the same plot. Note: the only output from the function is the plot.

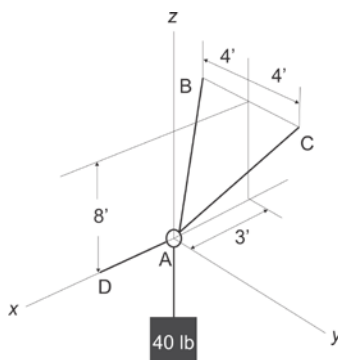
script.m

```
ang=linspace(0,365,100);  
hold on  
plotcos(ang)  
plot(ang,sind(ang),'yh')
```

plotcos.m

```
function plotcos( ang )  
%plot cosine  
plot(ang,cosd(ang),'m*')  
end
```

4. You are responsible for evaluating the forces in the cables shown below. Create a modular program that will ask the user for the following inputs: Coordinates of points B, C, D and the weight of the box. The output of this program will be the forces in each of the cables.



Script file

```
%inputs: coordinates of points, weight of box
%outputs: force in each cable
a=[0,0,0];

b=input('Enter the coordinates of point b in [x,y,z] format: \n');
c=input('Enter the coordinates of point c in [x,y,z] format: \n');
d=input('Enter the coordinates of point d in [x,y,z] format: \n');
weight=input('Enter the weight of the box:\n');

%call uv function to calculate the unit vectors
unit_ab=uv(a,b);
unit_ac=uv(a,c);
unit_ad=uv(a,d);

%call system_solve to solve for the force magnitudes in each cable
forces=system_solve(unit_ab,unit_ac,unit_ad,weight);

fprintf('The force in cable AB is %.5f lbs.\n',forces(1))
fprintf('The force in cable AC is %.5f lbs.\n',forces(2))
fprintf('The force in cable AD is %.5f lbs.\n',forces(3))
```

uv.m

```
function [ unit_v ] = uv( point_from,point_to )
%Calculate the unit vector for a cable between two given points

%call pos_vector function to get position vectors between two points
position_vec=pos_vector(point_from,point_to);
unit_v=position_vec/norm(position_vec);
end

function [r]=pos_vector(ptA,ptB)
%calculate the position vector between two points
%assume going from ptA to ptB

r=ptB-ptA;
end
```

system\_solve.m

```
function [ forces ] = system_solve(unit1,unit2, unit3, weight)
%Solve system of equations to determine the force magnitudes in each cable
%equation one: sum of x-components
%equation two: sum of y-components
%equation three: sum of z-components
%form of system of equations:  $Ax=b \dots x=(A^{-1}) * b$ 

A(:,1)=unit1;
A(:,2)=unit2;
A(:,3)=unit3;

b(3,1)=weight;
forces=inv(A)*b;
end
```

5. Create a script and function that will allow the user to enter three values and have the running average computed.

Script.m

```
x=input('Enter a number:\n');
average=persist(x)
x1=input('Enter a number:\n');
average=persist(x1)
x2=input('Enter a number:\n');
average=persist(x2)
```

persist.m

```
function [avg] = persist(x)
%find the average of the numbers entered
persistent n
persistent sumx
if isempty(n)
    n=0;
end
if isempty(sumx)
    sumx=0;
end
n=n+1;
sumx=sumx+x;
avg=sumx/n;
end
```

6. You can also make a variable **global** – meaning it can be used by all functions/scripts. Use the kinematic equation to calculate displacement in the y-direction. Make  $g = 9.81$  a global variable.

$$y = y_o + v_o t - 0.5gt^2$$

Script file

```
global g
g=9.81; % acceleration due to gravity
yo=0; %initial displacement
vo=200; %initial velocity
t=20; %time
position=displace(yo,vo,t);
fprintf('The position is %.4f meters.\n',position)
```

displace.m

```
function [ y ] = displace( yo,vo,t )
%find y position
global g
y=yo+vo*t-.5*g*t^2;
end
```

